Project 2

```
library(mosaic)
```

INTRODUCTION

Health insurance is a contract that requires an insurer to pay some or all of a person's healthcare costs in exchange for a premium. More specifically, health insurance typically pays for medical, surgical, prescription drug, and sometimes dental expenses incurred by the insured. Health insurance can reimburse the insured for expenses incurred from illness or injury, or pay the care provider directly. Individual health insurance health policies are by far the fastest growing form of U.S. health insurance. To develop the best medical insurance products, the insurer need access to historical data to approximate the medical costs of each user. With this data, a medical insurer can develop more accurate pricing models, plan a particular insurance outcome, or manage a big portfolios. For all these cases, the objective is to accurately predict insurance costs.

QUESTION

1.How do age and sex impact the insurance charge?

2.What is the relation betwwen age and sex on insurance charge?

3.Predicting insurance charges based on sex and their age

THE DATA

This Data is a pratical is used in the book Machine Learning with R by Brett Lantz; which is a book that provides an introduction to machine learning using R. All of these datasets are in the public domain but simply needed some cleaning up and recoding to match the format in the book. The following data obtained from Kaggle, explain the cost of a small sample of USA population Medical Insurance Cost based on some attributes depicted on "Content".

The aim of the dataset is to find the correlation between age and health insurance charges, between sex and health insurance charges, among age, sex and health insurance charges. The dataset contains 3 variables:

age: quantiative: age of primary beneficiary

sex: categorical: insurance contractor gender, female, male

charges: quantiative: Individual medical costs billed by health insurance

CONCLUSION

As we could see, age and sex don't have a big impact on the insurance charges. Cause the we could spot that there are quite a lot of people who have the insurance charges much higher than the model charge. There are 3 types of insurance charges that we don't have information yet: 1 is from 0 to 10000, $2 is from 10000$ to 30000$ and 3 is above 30000$. We still need more data to have a better understanding about this dataset.

```
pj2 = read.csv("insurance.csv")
head(pj2)
```

```
##   age    sex   charges
## 1  19 female 16884.924
## 2  18   male  1725.552
## 3  28   male  4449.462
## 4  33   male 21984.471
## 5  32   male  3866.855
## 6  31 female  3756.622
```

ANALYSIS

1.

This dataset contains 1338 people.

```
nrow(pj2)
```

```
## [1] 1338
```

The average insurance charge of one person is 13270.42$

```
mean(charges~1,data=pj2)
```

```
##        1
## 13270.42
```

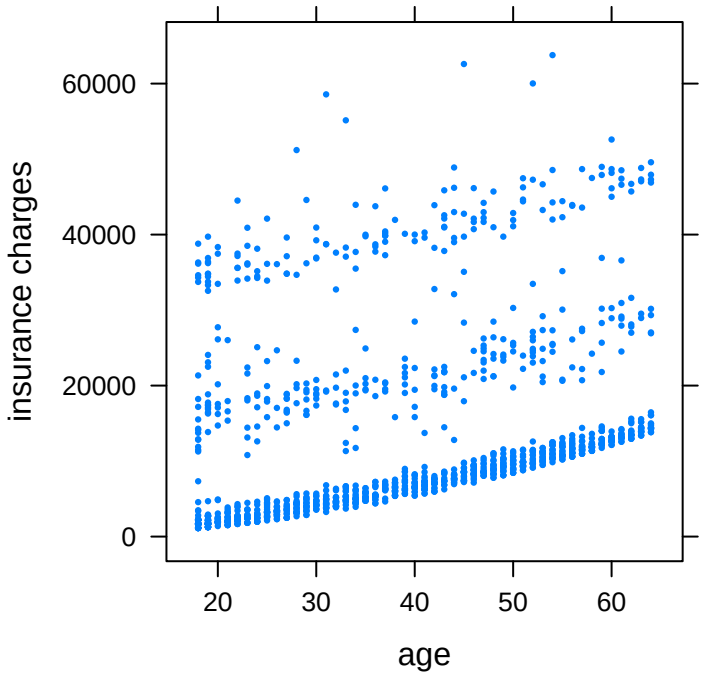
The average insurance charge of female is 12569.58$ compared to 13956.75$ of male.

```
mean(charges~sex,data=pj2)
```

```
##   female     male
## 12569.58 13956.75
```

First, we'll have to see the relation between insurance charges and age by using a scatterplot, where the explanatory variable is age and the response variable is the inusrance charges

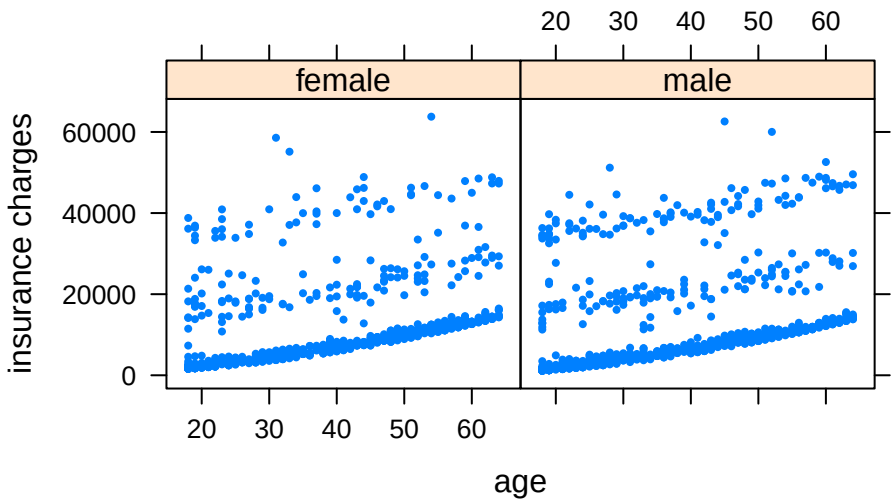
```
xyplot(charges~age,data=pj2,xlab = "age",ylab="insurance charges",pch=16,cex=0.4)
```



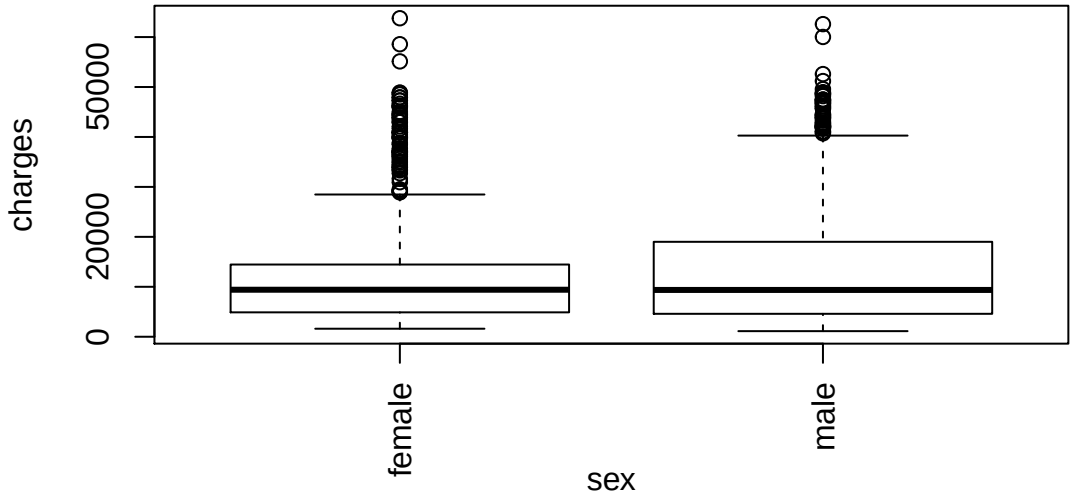
We could see that the older people get, the more expensive the insurance charges are.

2. Now we we use scatterplot again to find the relation between male and female's insurance charges.

```
xyplot(charges~age | sex, data=pj2,cex=0.5,pch=16,
       xlab = "age",ylab="insurance charges")
```




```
boxplot(charges~sex,data=pj2, las=3 )
```



We cannot find much difference between cost paid by male and female in first plot and sex is alomst equally scatteret.The bar chart also shows that there is very little difference between average cost paid by male and female. Hence, sex of the person does not necessarily determine the insurance charges one pays.

3.

```
mod1=lm(charges~1,data=pj2)
mod2=lm(charges~sex,data=pj2)
mod3=lm(charges~age,data=pj2)
mod4=lm(charges~sex+age,data=pj2)
mod5=lm(charges~sex+age+sex*age,data=pj2)
```

The first model is the simple "everyone has the same insurance charge" model.

insurance charge = const + random

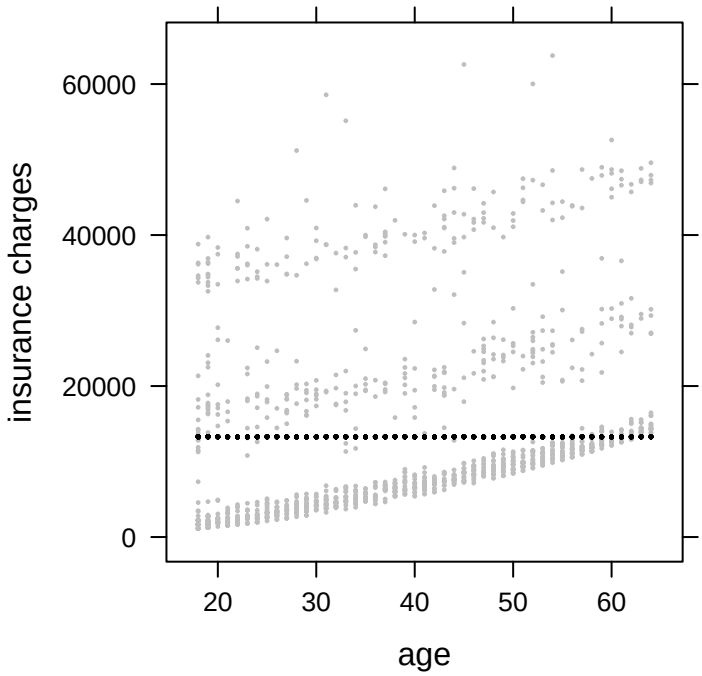
```
coef(mod1)
```

```
## (Intercept)
##    13270.42
```

```
mod1
```

```
##
## Call:
## lm(formula = charges ~ 1, data = pj2)
##
## Coefficients:
## (Intercept)
##       13270
```

```
xyplot(charges+fitted(mod1)~age,data=pj2,
       xlab = "age",ylab="insurance charges",
       col=c("grey","black"),cex=0.3,pch=16)
```



The mean insurance charge of female vs male: categorical

insurance charge ~ sex = const + sex + random factor

```
coef(mod2)
```

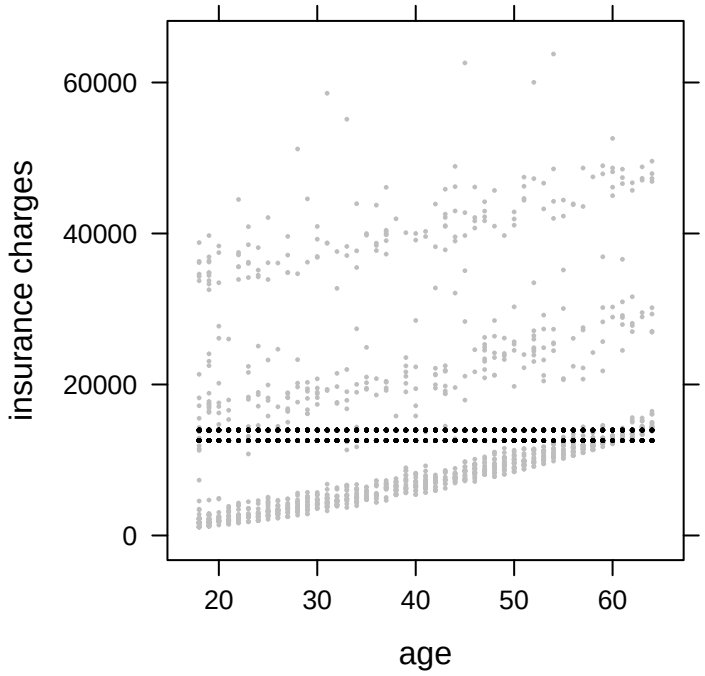
```
## (Intercept)     sexmale
##   12569.579    1387.172
```

```
mod2
```

```
##
```

```
## Call:
## lm(formula = charges ~ sex, data = pj2)
##
## Coefficients:
## (Intercept)      sexmale
##       12570         1387
```

```
xyplot(charges+fitted(mod2)~age,data=pj2,
       xlab = "age",ylab="insurance charges",
       col=c("grey","black"),cex=0.3,pch=16)
```



We can notice that older people tend to pay slightly more but to make it more clearer we can draw a regression link. The regression line shows positive correlation. Hence, age plays small role in predicting insurance price.

insurance charge ~ age = const + cofficient x age + random

```
coef(mod3)
```

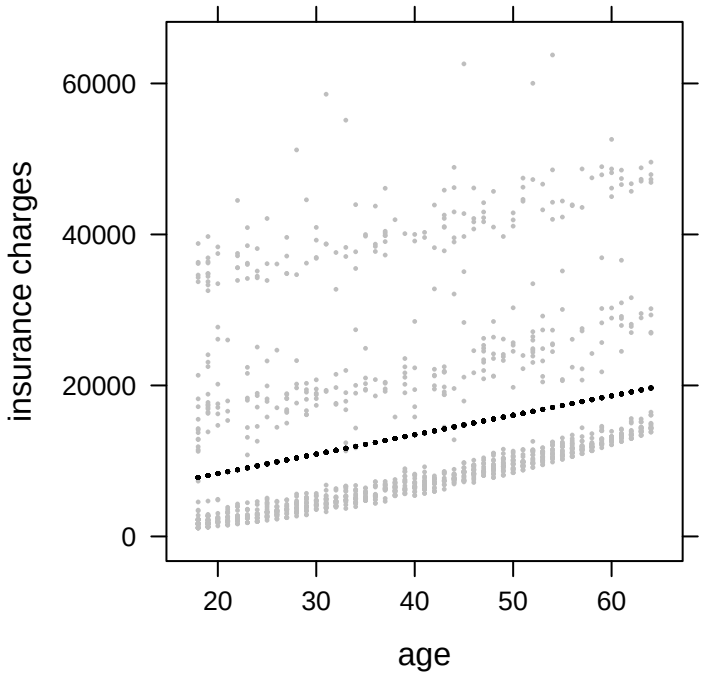
```
## (Intercept)         age
##   3165.8850    257.7226
```

```
mod3
```

```
##
## Call:
## lm(formula = charges ~ age, data = pj2)
##
## Coefficients:
## (Intercept)          age
##      3165.9        257.7
```

```
xyplot(charges+fitted(mod3)~age,data=pj2,
       xlab = "age",ylab="insurance charges",
```

```
       col=c("grey","black"),cex=0.3,pch=16)
```



When people get older, they receive more expensive insurance charge.

In this model 4 we predict the correlation of betwwen both sex and age and insurance charge

insurance charge ~ sex + age = const + sex + cofficient x age + random factor

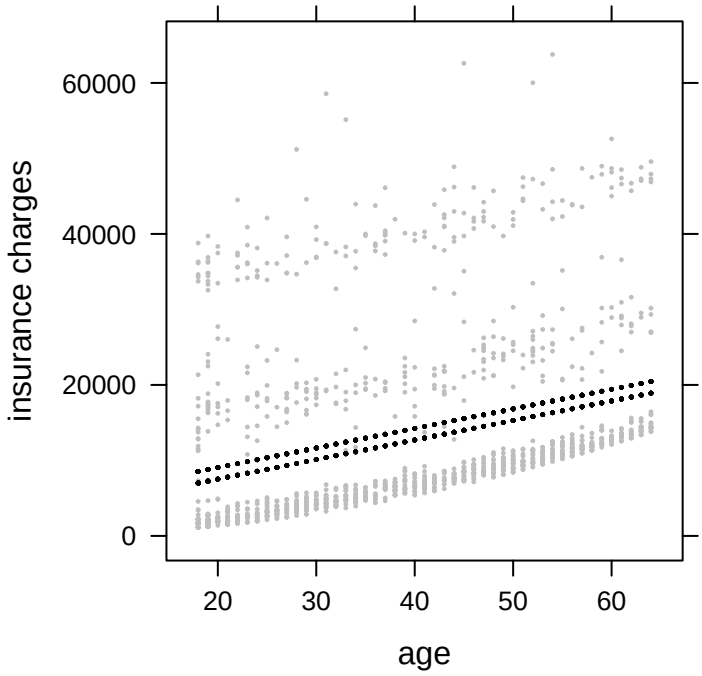
```
coef(mod4)
```

```
## (Intercept)     sexmale         age
##   2343.6249   1538.8314    258.8651
```

```
mod4
```

```
##
## Call:
## lm(formula = charges ~ sex + age, data = pj2)
##
## Coefficients:
## (Intercept)      sexmale          age
##      2343.6       1538.8        258.9
```

```
xyplot(charges+fitted(mod4)~age,data=pj2,
       xlab = "age",ylab="insurance charges",
       col=c("grey","black"),cex=0.3,pch=16)
```



The 5th model indicates the interaction between 2 variables: age and sex. And we couldn't see the interaction between sex and age.

insurance charge ~ sex * age = const + sex + cofficient x age + sex:age

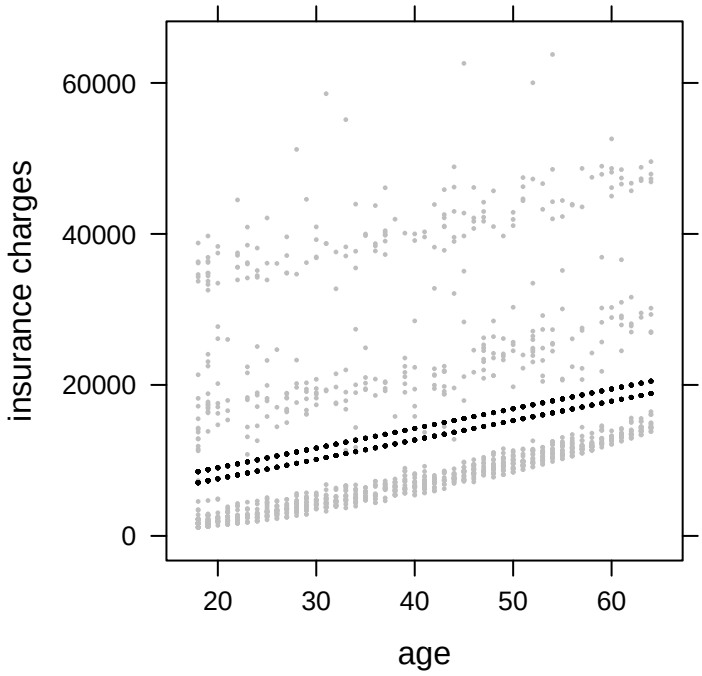
```
coef(mod5)
```

```
## (Intercept)     sexmale         age sexmale:age
## 2416.848522 1394.925331  257.011490    3.669849
```

```
mod5
```

```
##
## Call:
## lm(formula = charges ~ sex + age + sex * age, data = pj2)
##
## Coefficients:
## (Intercept)      sexmale          age  sexmale:age
##     2416.85      1394.93       257.01         3.67
```

```
xyplot(charges+fitted(mod5)~age,data=pj2,
       xlab = "age",ylab="insurance charges",
       col=c("grey","black"),cex=0.3,pch=16)
```


insurance charges
60000
40000
20000
0
20
30
40
50
60
age